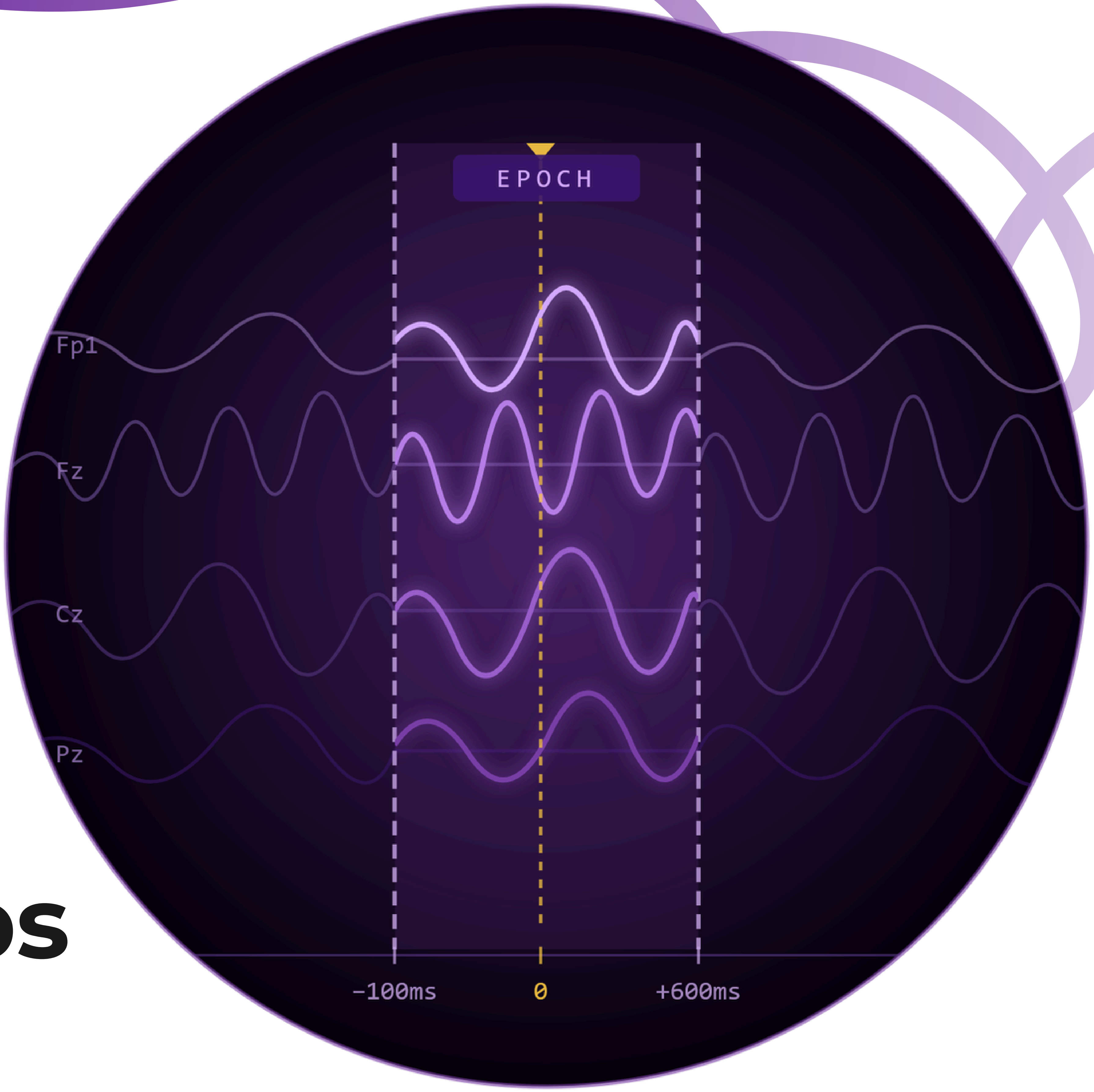
Student
Network for
Neuroscience

PsyFaKo

MES
FORSCHUNGSSYSTEME GMBH
Lösungen für neurowissenschaftliche Anwendungen

FACHSCHAFT PSYCHOLOGIE · HUMBOLDT-UNIVERSITÄT ZU BERLIN ·



EEG Preprocessing & Analysis - First steps

Contents

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What is EEG?

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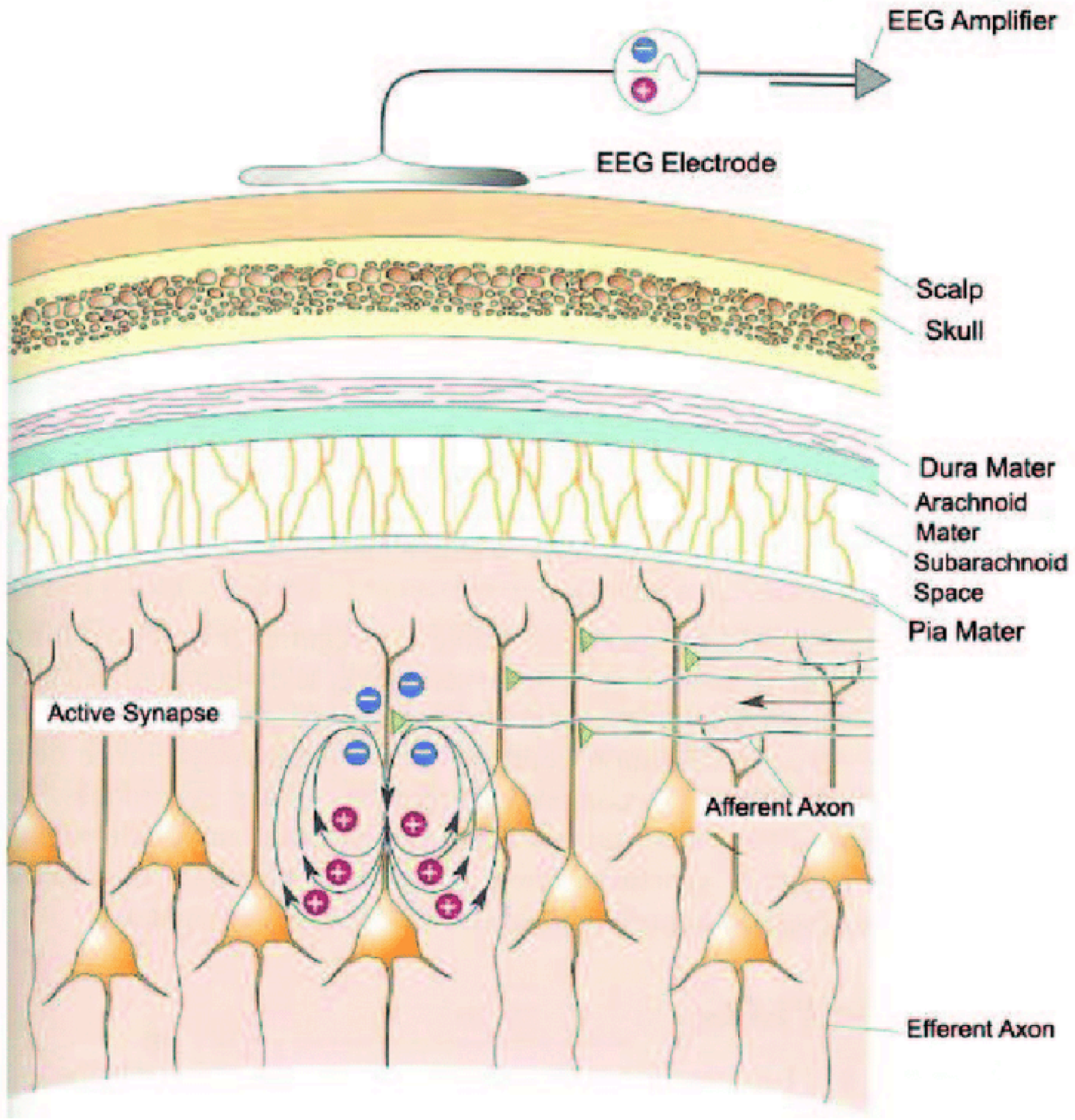
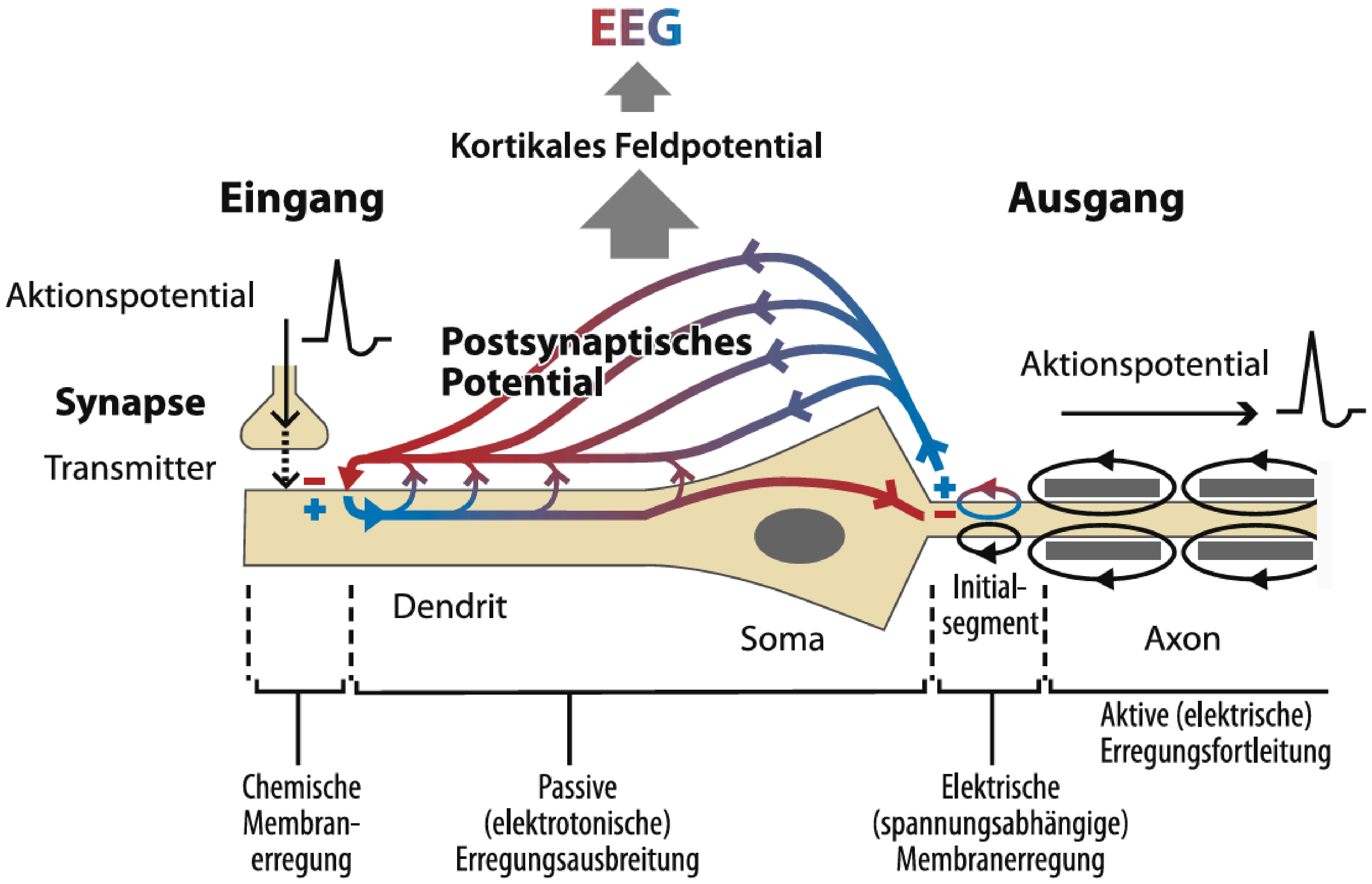
- Electroencephalography (EEG)
- one of the most important non-invasive methods for studying human cognitive function and diagnosing brain diseases
- mostly measurement of **synaptic** activity of layer 5 pyramidal neurons
- electrodes measure voltage-**differences**(µV)

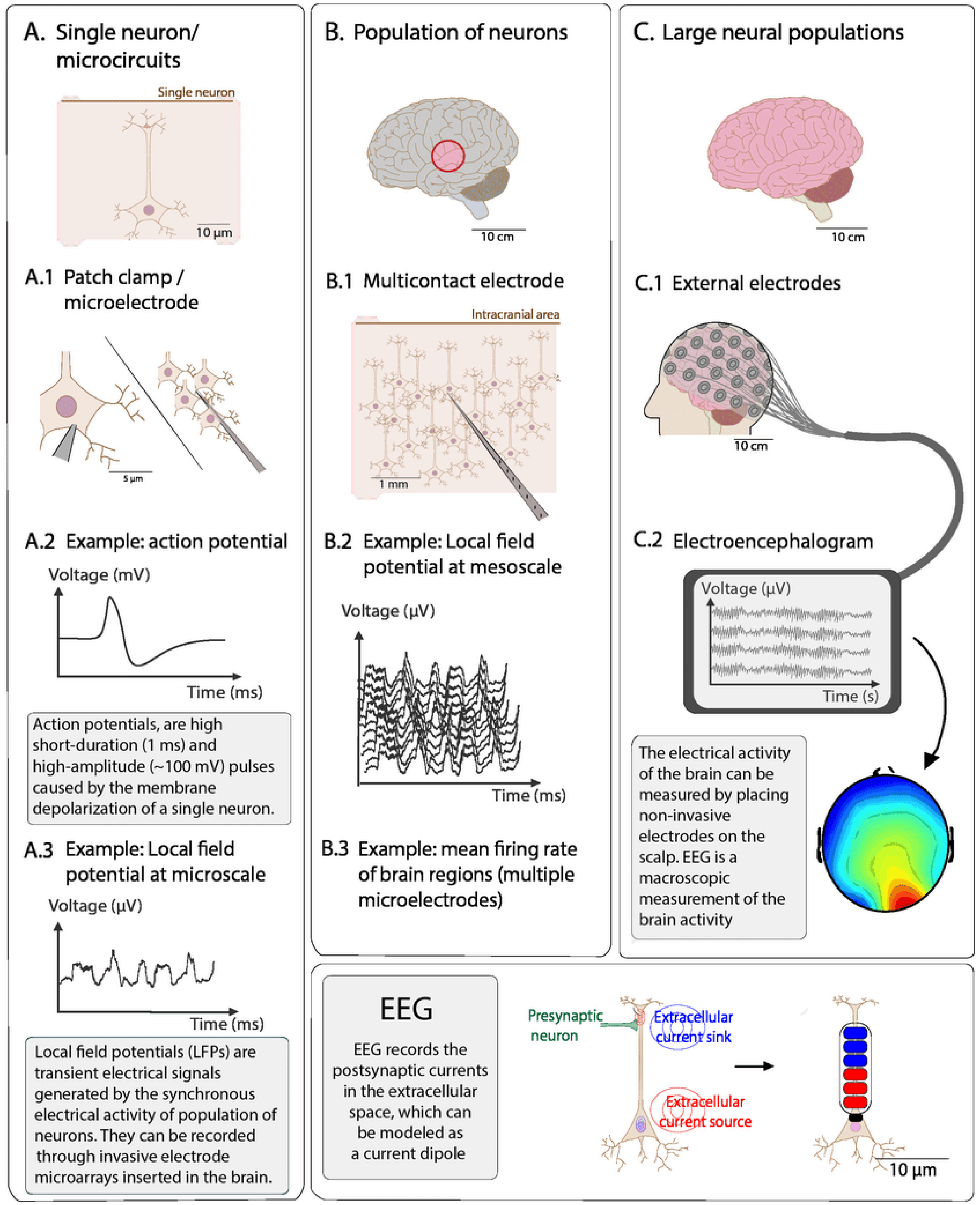


Uldry & Millán, 2007 – Brain–Machine Interface study

What is EEG?

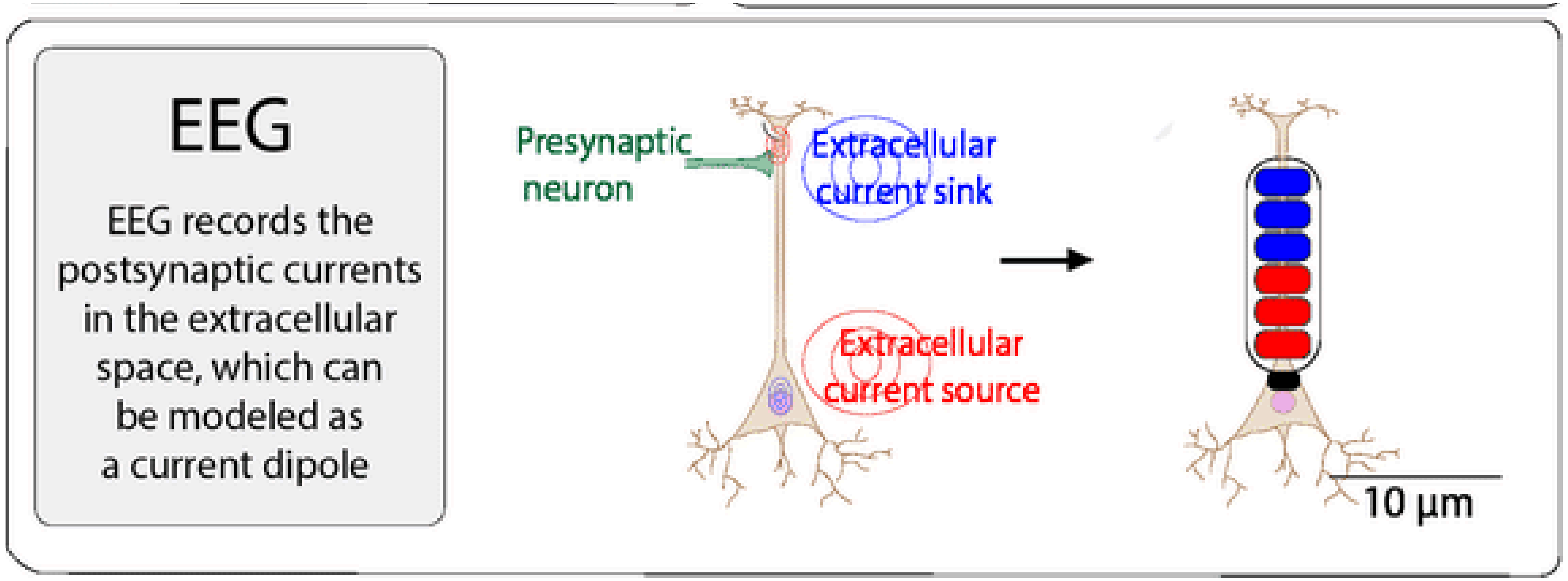


Zschocke, S., & Hansen, H. (2023). .

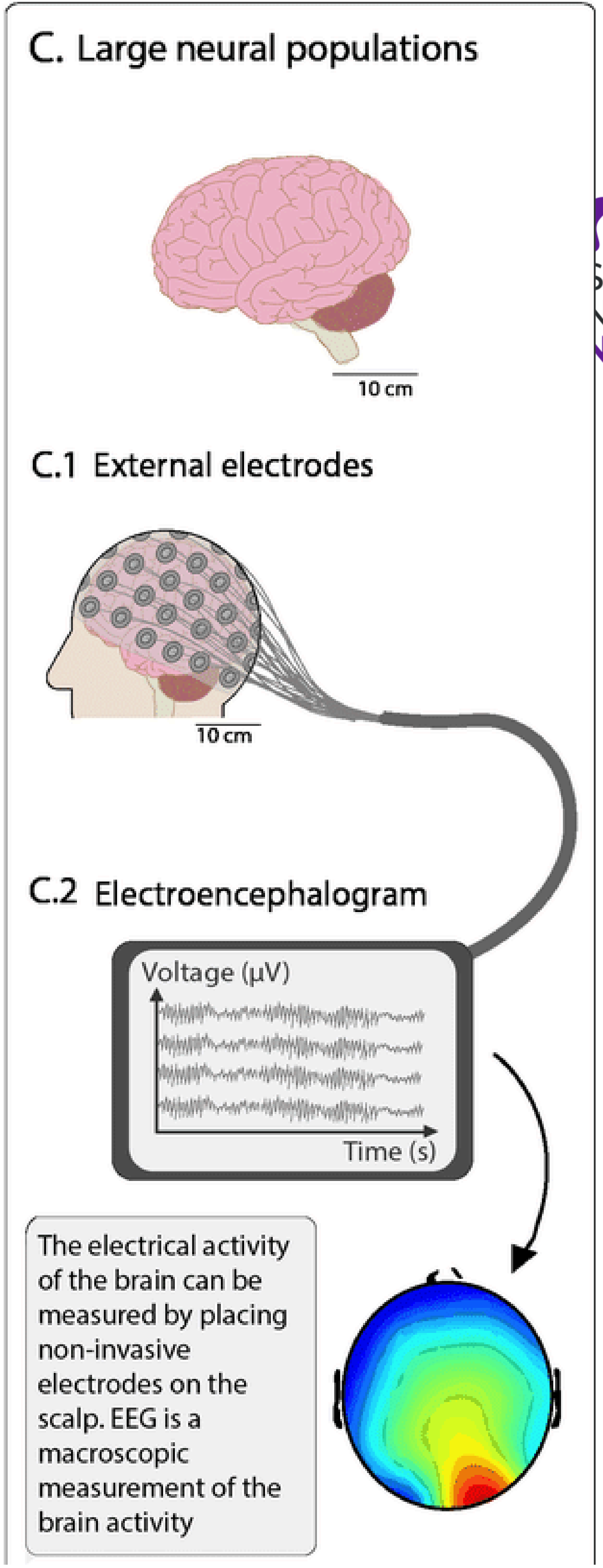


Computational models in Electroencephalography, Glomb, Cabral, Cattani and Mazzoni; September 2020

What is EEG?



Computational models in Electroencephalography, Glomb, Cabral, Cattani and Mazzoni; September 2020



What is EEG?

- signal from each recording electrode is measured as a voltage **difference relative** to some reference:
  - (a) single electrode
  - (b) mastoids (M1/M2/Linked)
  - (c) average of electrodes

- ground electrode: typically used to establish a common electrical potential for the recording system

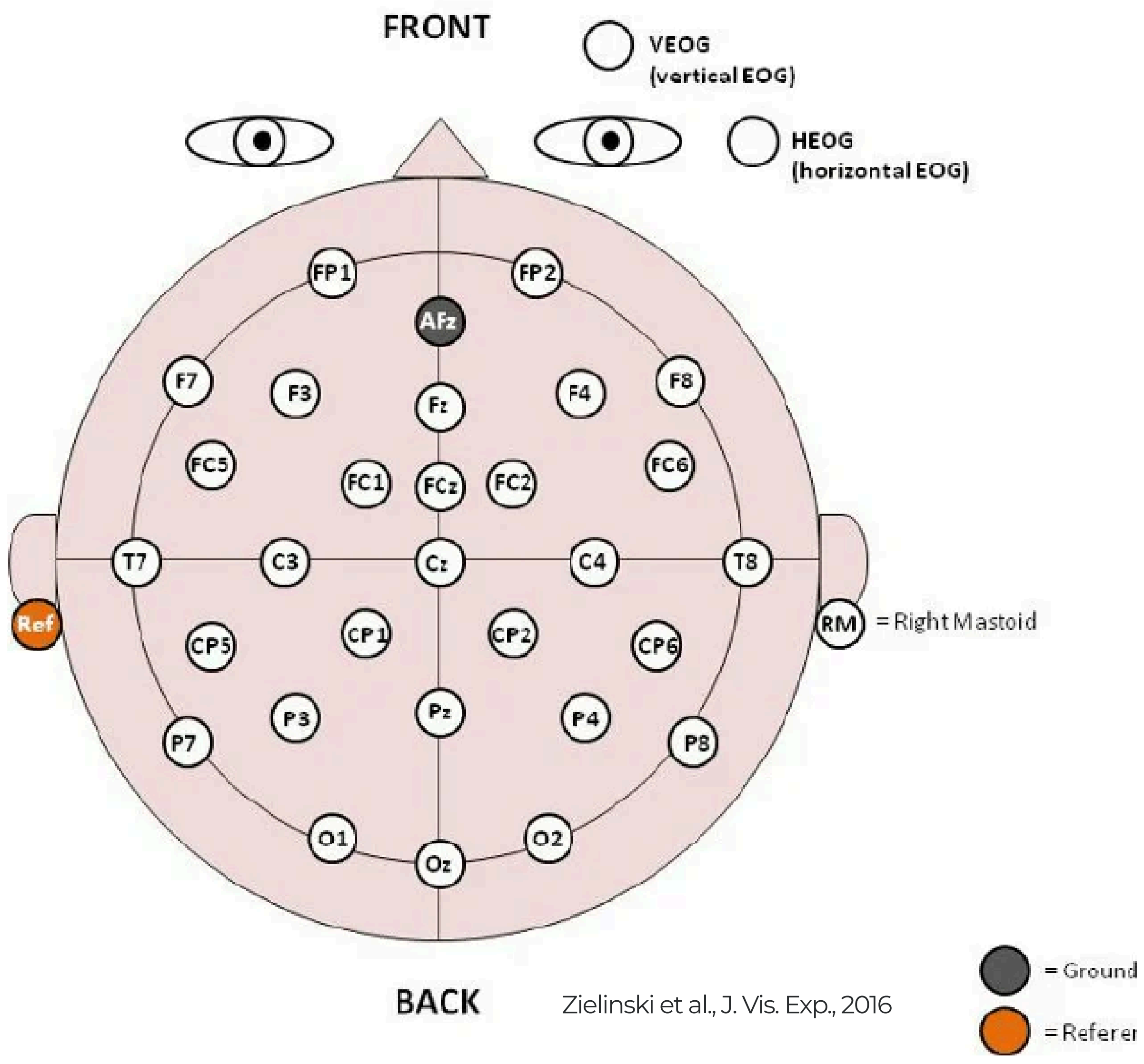
→ minimizes common mode noise

- **Re-referencing** in EEG data analysis: raw recordings with a single reference often contain significant distortions

→ Goal: estimate a more neutral reference point to reduce these effects

How do reference montage and electrodes setup affect the measured scalp EEG potentials? Hu et. al, 2018

$$\mathbf{v}_r = \mathbf{T} \cdot \mathbf{v}_\infty$$



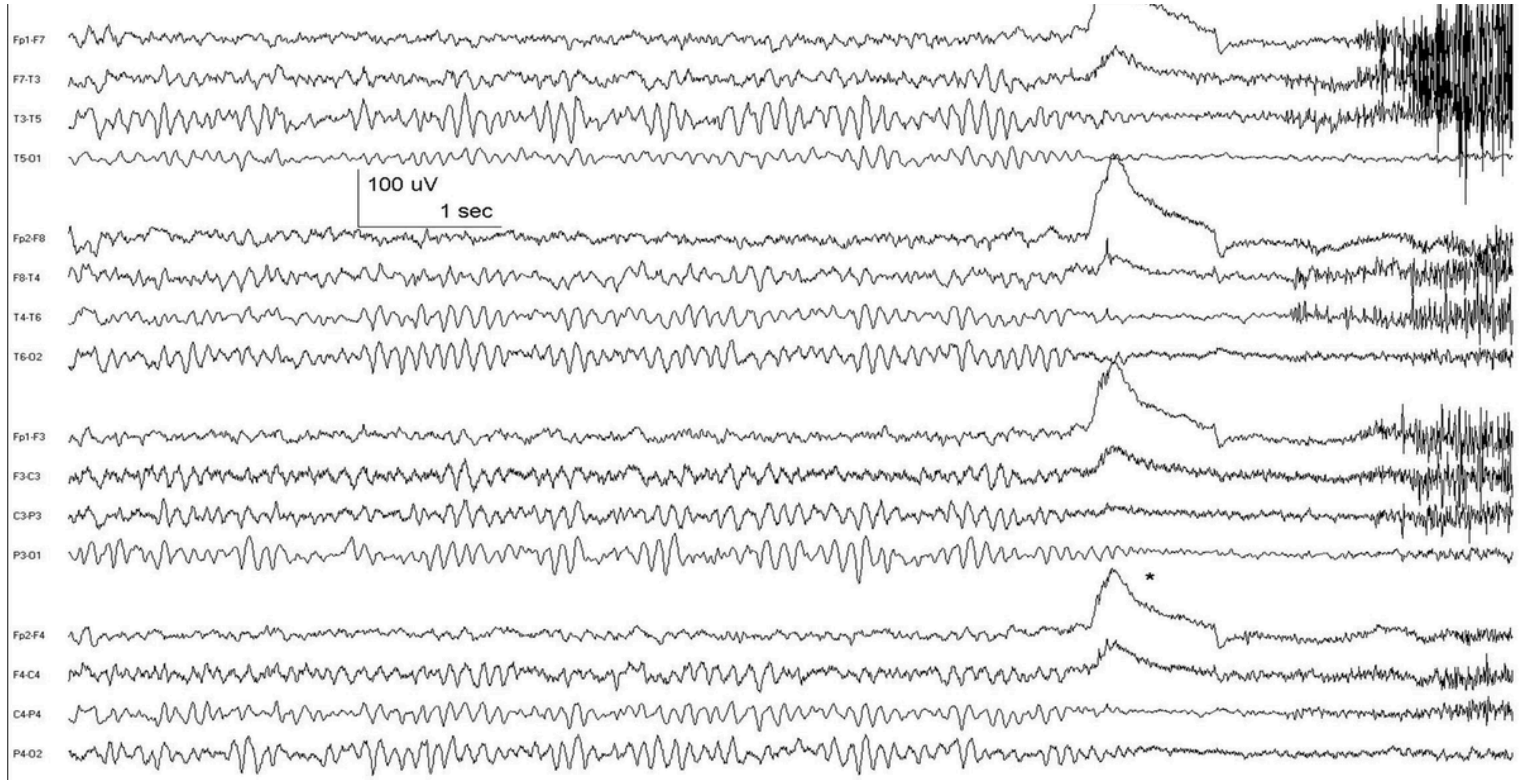
Zielinski et al., J. Vis. Exp., 2016

What is EEG?

EEG Data Format

- continuous electrical activity recorded from the scalp over time
- multichannel time series
  - Channels (Electrodes): Each electrode records the electrical potential **difference** to a reference electrode
  - Time Points
  - Amplitude Values
- Associated Metadata
  - Sampling Frequency (Hz)
  - Channel Names and Types
  - Electrode Locations
  - Subject Information
  - Experimental Conditions
  - Event Markers/Annotations

awake, resting; normal posterior alpha rhythm disappears with eye opening (*); high frequency activity after eye opening is muscle artifact; anterior-posterior bipolar montage; 0.5 -70 Hz



What is EEG?

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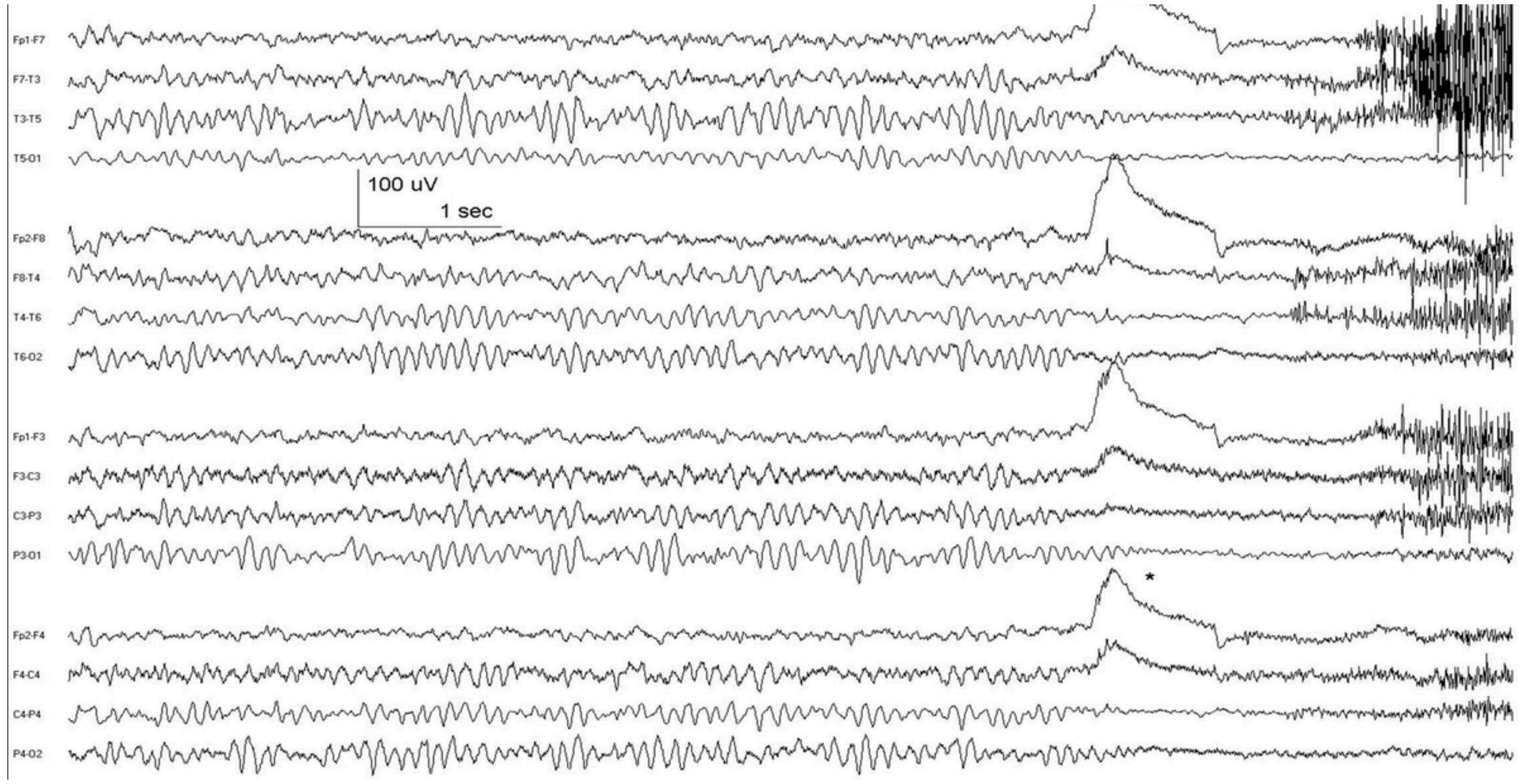
EEG Data Format

- Common File Formats
  - EDF (European Data Format) / BDF (BioSemi Data Format)
  - BrainVision Core Data Format (VHDR/VMRK/EEG)
  - FIF (MEG/EEG data format)
  - EEG hardware manufacturers' own specific file format
- Standardized Data Structures (BIDS)
  - sub- folders
  - ses- folders
  - eeg- subfolders
  - _events.tsv files

awake, resting; normal posterior alpha rhythm disappears with eye opening (*); high frequency activity after eye opening is muscle artifact; anterior-posterior bipolar montage; 0.5 -70 Hz



What is EEG?

How to access the data

- Usually: Some sort of programming environment (e.g. Python)
- MNE-Python: open-source software package
- But also: User-Friendly Interfaces
  - tools like EEG-Pype provide a graphical user interface (GUI) built on top of MNE-Python functions
  - Similarly, other toolboxes like osl-ephys also build upon MNE-Python
  - toolboxes like EEGLAB provide an alternative standalone processing package

```
import os

import numpy as np

import mne

sample_data_folder =
mne.datasets.sample.data_path()
sample_data_raw_file = os.path.join(
    sample_data_folder, "MEG", "sample",
"sample_audvis_raw.fif"
)
raw =
mne.io.read_raw_fif(sample_data_raw_file)
raw.load_data()
```

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Pipeline

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Load & Merge → Filter → Extract Epochs → Baseline → Artifact Corr. → Average

Load Data

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Used file format: BrainVision Core Data Format

- **.vhdr:** Header file, metadata about EEG recording; text-based file with information (sampling rate, number of channels, channel names, electrode impedances,... and **references** to corresponding .vmrk and .eeg files)
  → primary entry point for software
- **.vmrk:** Marker file, stores event markers or annotations for events/stimuli; essential for segmenting the continous EEG data into epochs
- **.eeg:** Data file; contains the raw, continuous EEG time-series data; typically a binary file that stores the actual voltage values recorded from each electrode at each time point

Filtering

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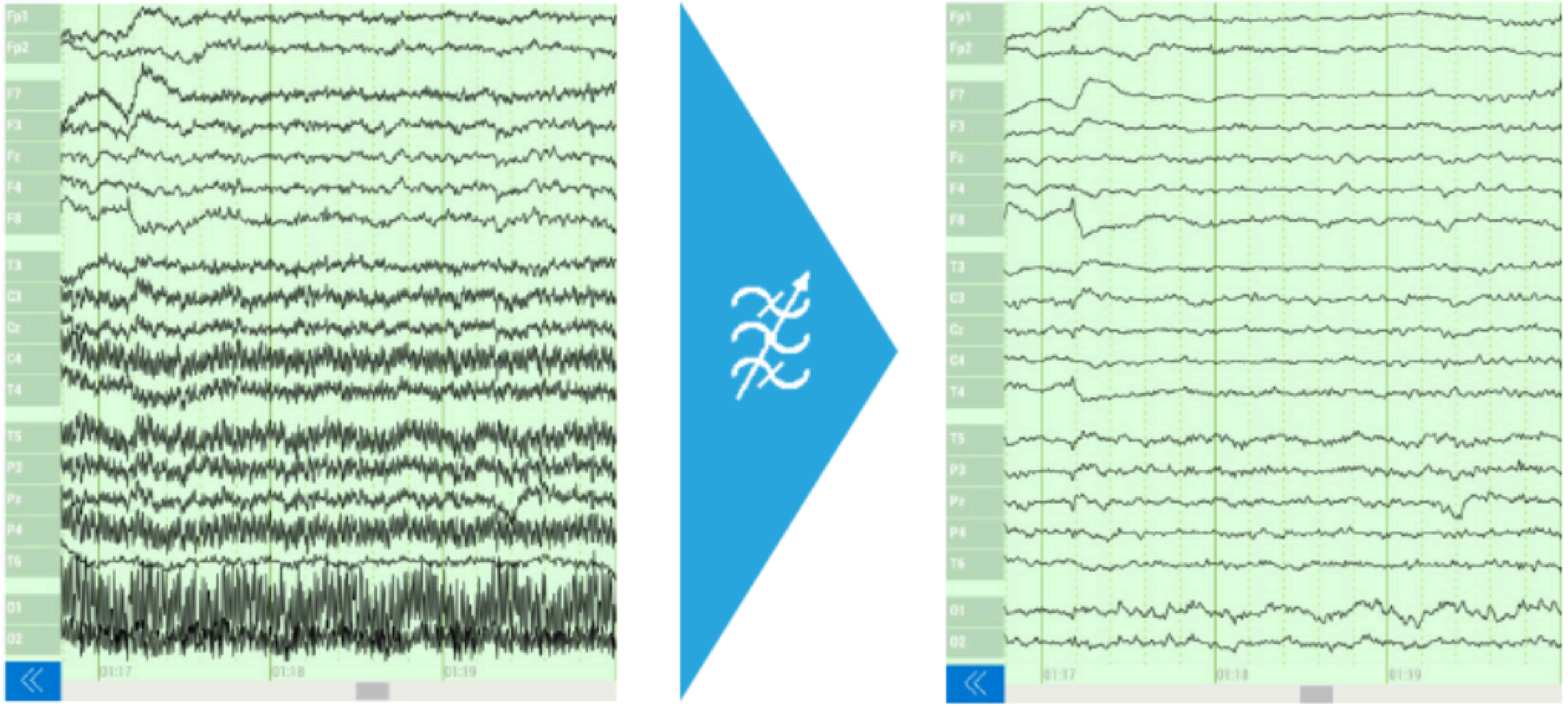
Filtering removes or attenuates parts of a signal. Specifically it attenuates frequencies.



Unfiltered EEG Signal

Filtered EEG Signal

https://zeto-inc.com/blog/eeg-signal-enhancement-digital-eeg-filters/#:~:text=Three%20main%20types%20of%20filters%20used%20to%20accomplish,also%20called%20notch%20filters%29%20%E2%80%93%20See%20Figure%201.

Filtering

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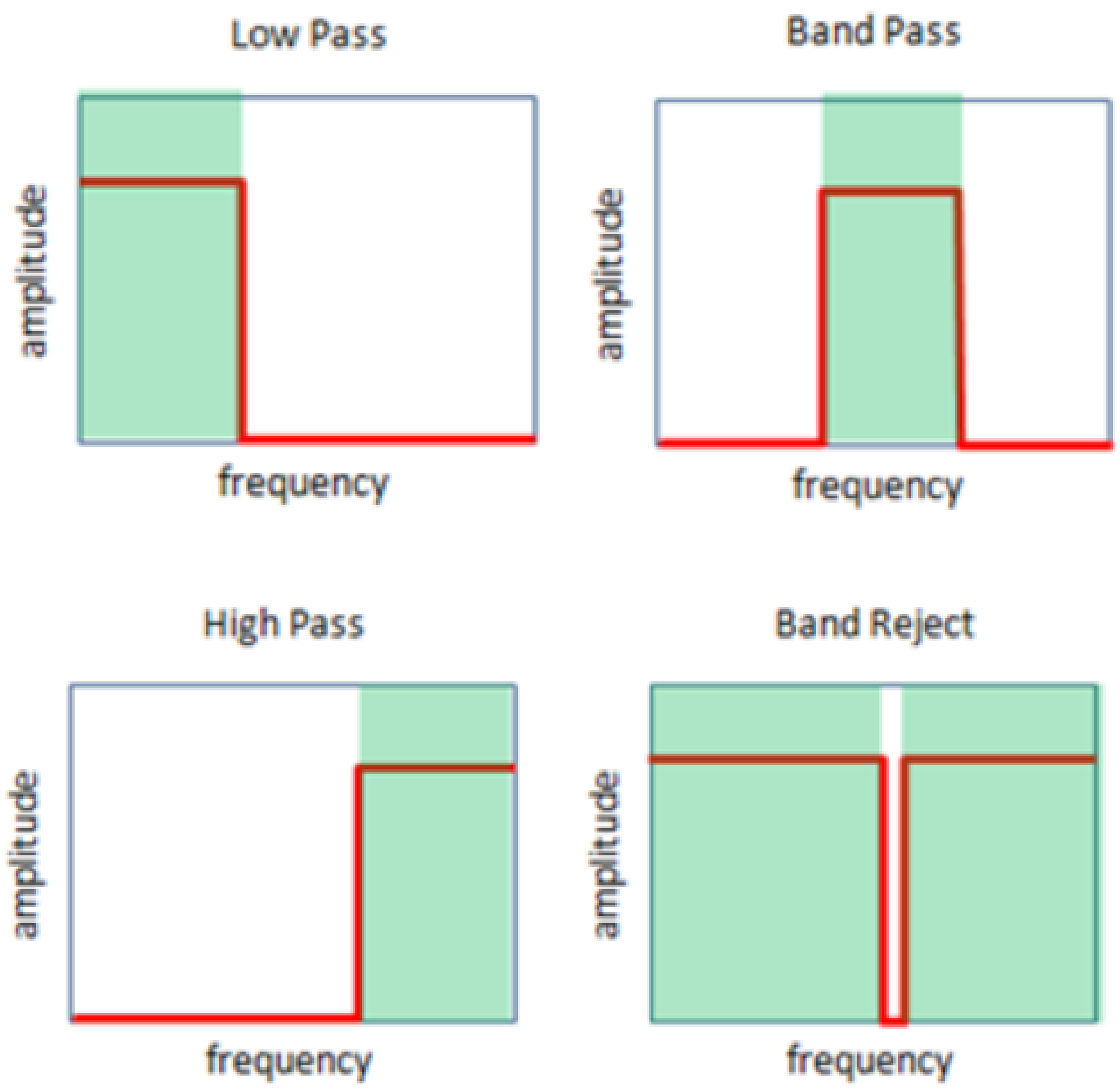
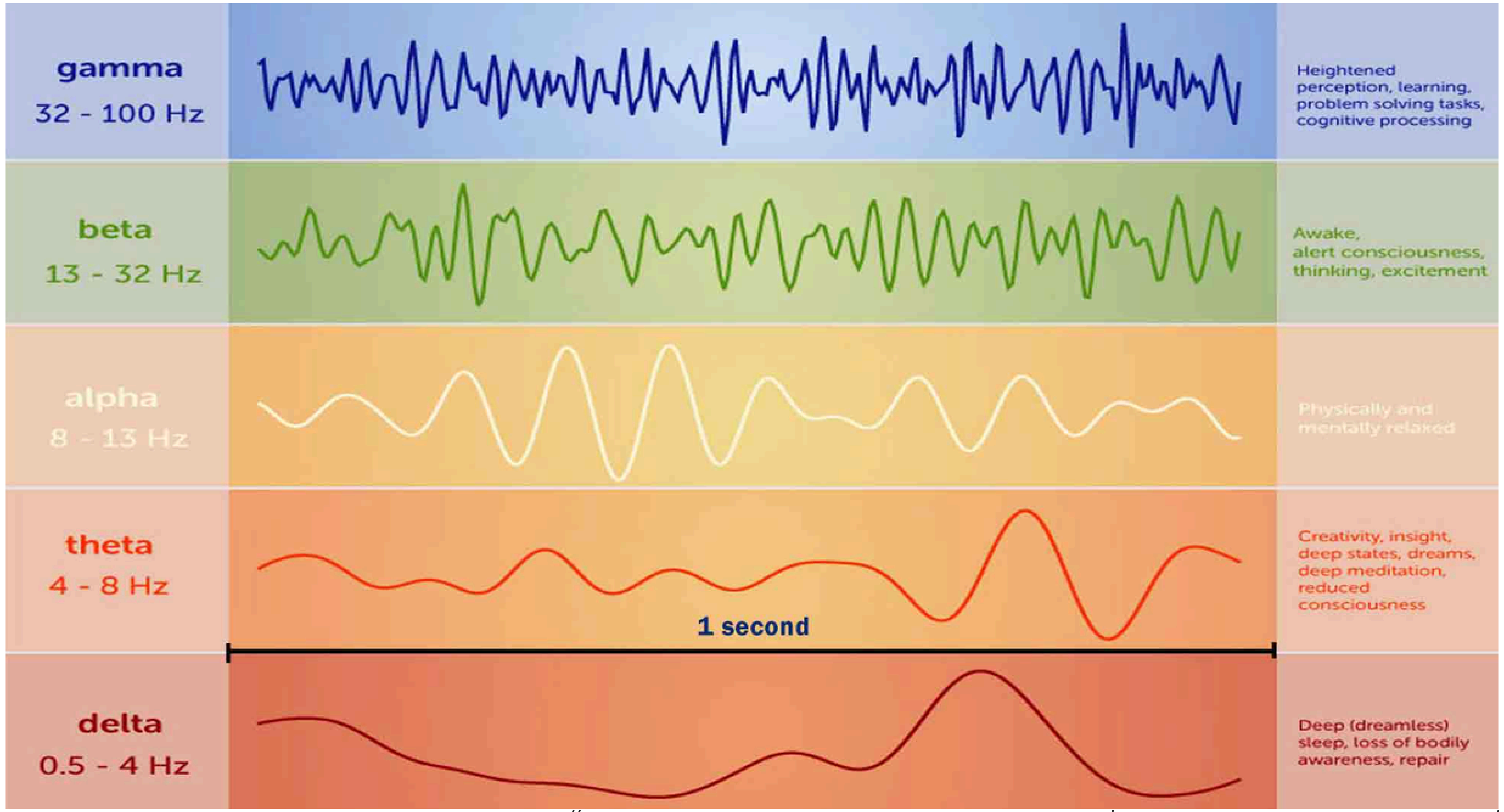
There are many types of filters that only remove pre-specified frequencies.



- filter types can help with the removal of typical recording artifacts
- mainly frequencies that are not reflecting neuronal oscillations
- Reminder: 1 Hz = one cycle per second

Filtering

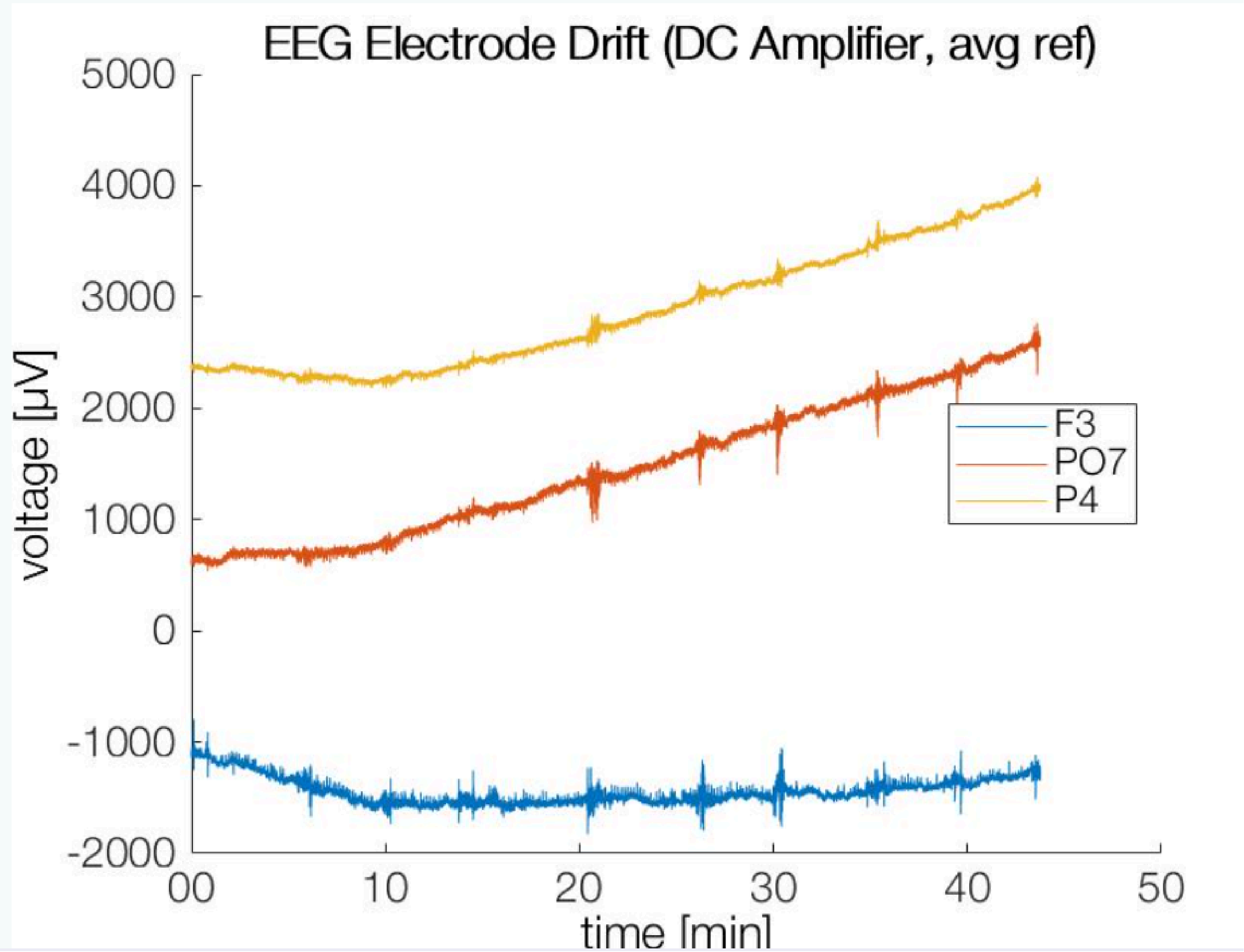
Not all oscillations are originating from the brain.
And from the ones that are, not all might be relevant for your paradigm.

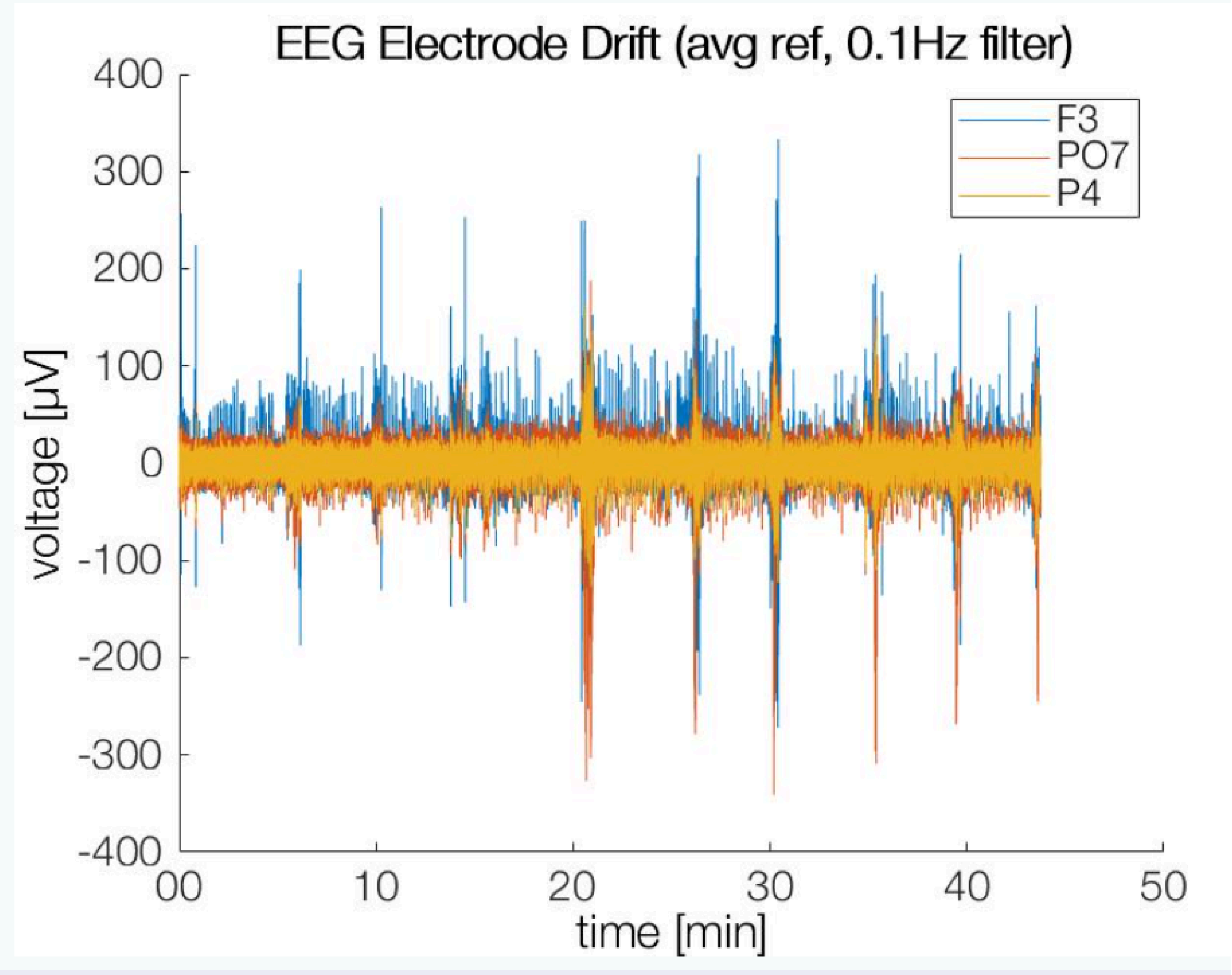


https://biofeedback-neurofeedback-therapy.com/brainwaves-in-neurofeedback/

Filtering

Filtering of low frequency drifts (e.g. sweat artifacts))

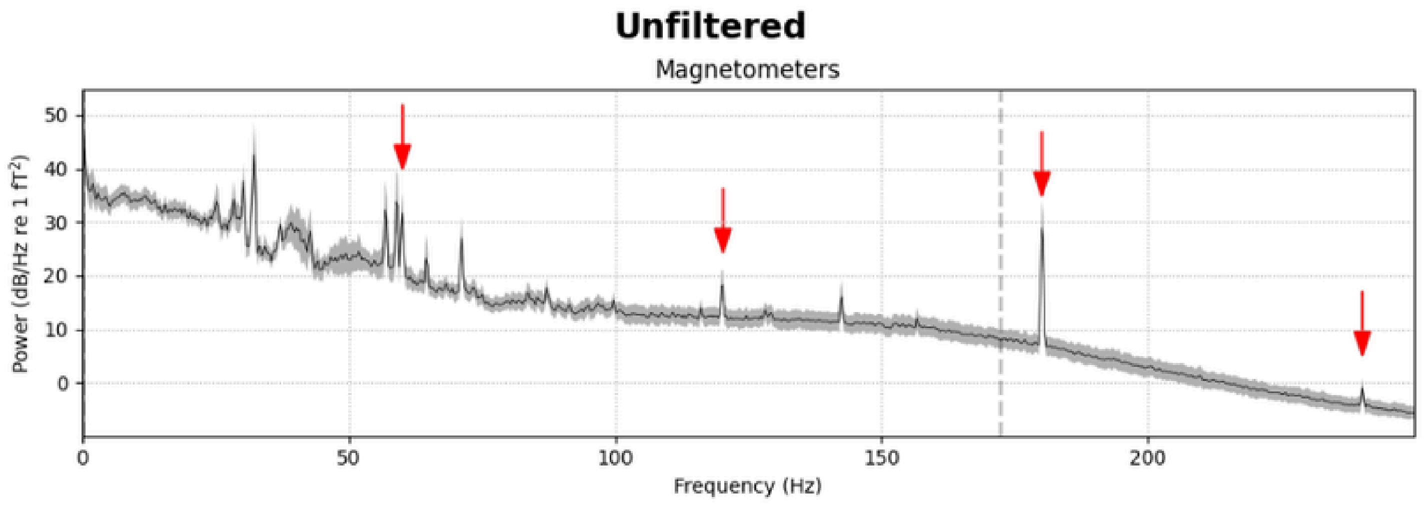


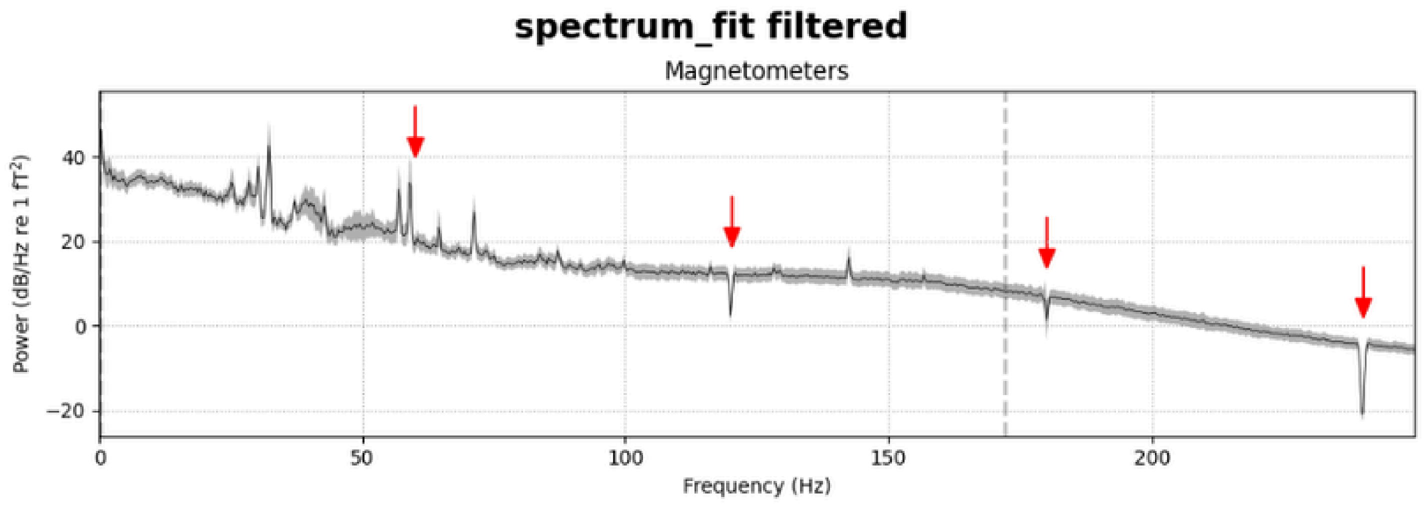


https://benediktehinger.de/blog/science/electrode-drift-in-eeg/

Filtering

Filtering of power line noise





https://mne.tools/stable/auto_tutorials/preprocessing/30_filtering_resampling.html

- power line noise mainly in MEG, since EEG is mostly done in shielded chambers
- frequencies:
  - 50 Hz in Europe
  - 60 Hz in the US
  - other countries may differ
- also affects all the respective harmonic frequencies

- There is so much more to filtering…but this is just to illustrate the concept!

Resampling

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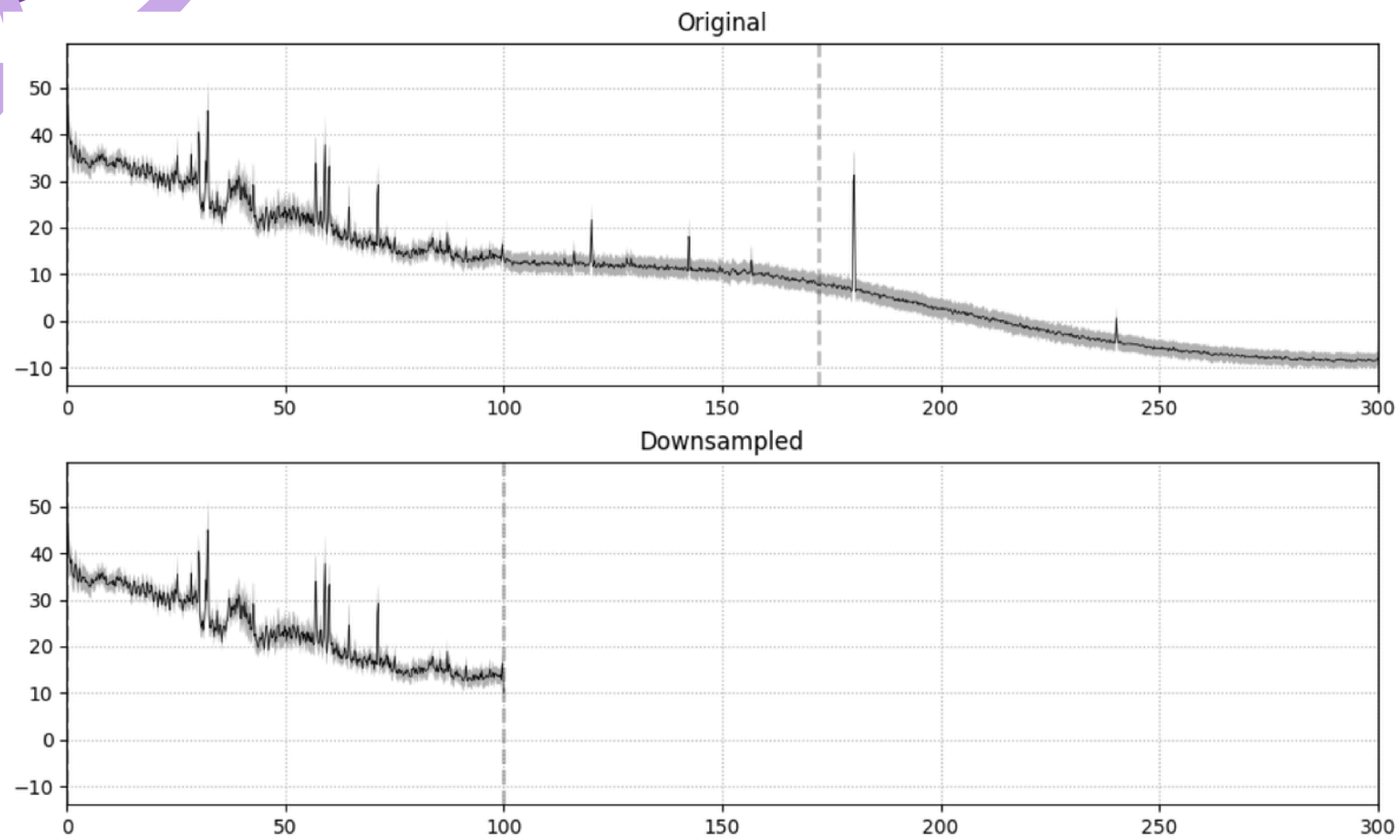
Works similarly to filtering, but completely removes specified frequencies.



- Downsampling saves computation time and memory space
- no information is lost, but timing of events can shift slightly
  - therefore event codes **MUST** be resampled as well to ensure precise timing
- resampling incorporates a low-pass filter per default to avoid aliasing
  - aliasing = overfolding of higher frequencies into lower ones during resampling
- double filtering is avoided if we used a smaller low-pass filter before

https://mne.tools/stable/auto_tutorials/preprocessing/30_filtering_resampling.html

Fragen?

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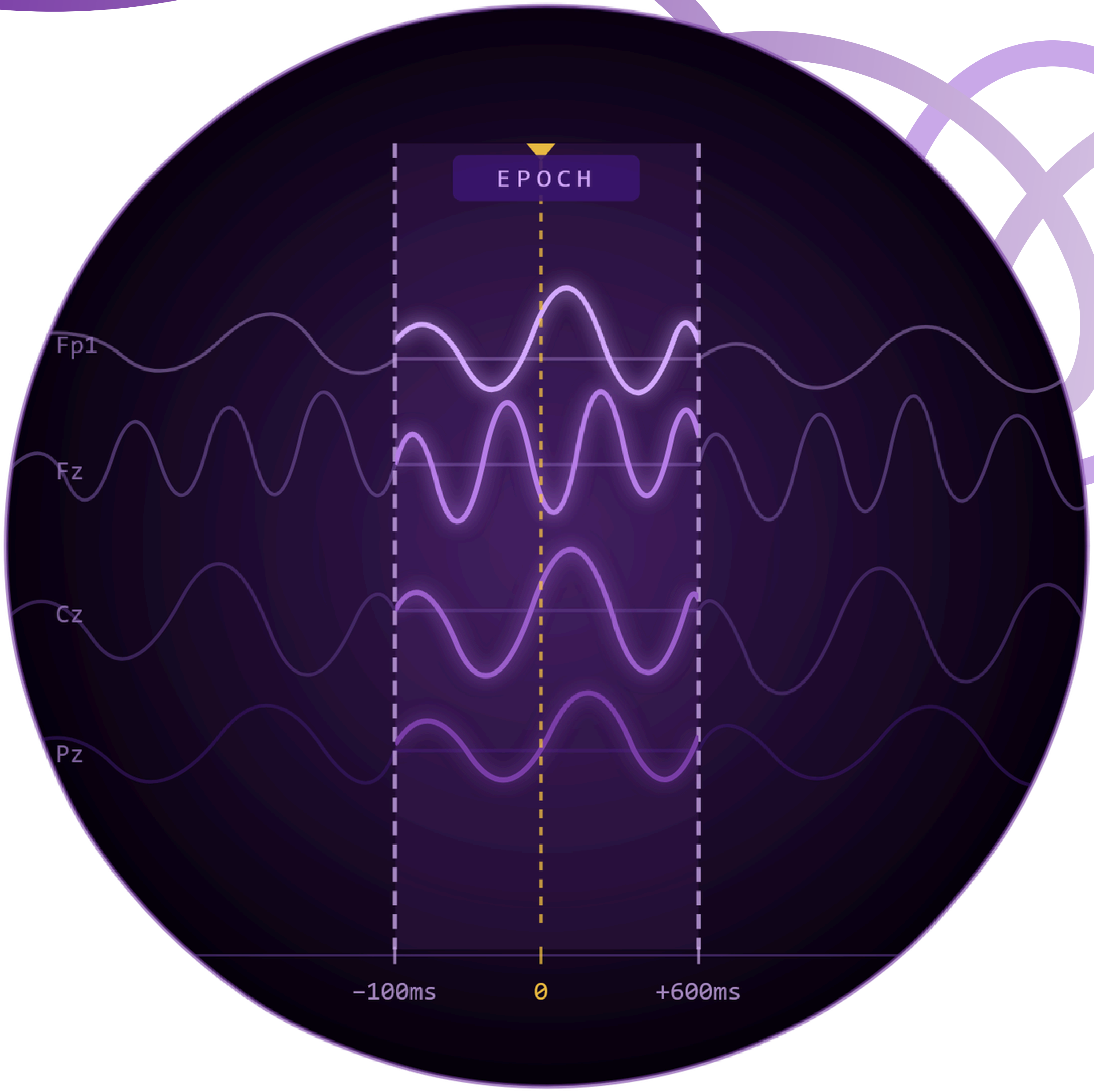
Epochs, Baseline & Artifact Correction



Re-referencing & Channel rejection

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Why do we need it?

Every EEG signal is a voltage difference relative to a reference electrode. The online reference (set during recording) can bias the signal. We re-reference to the average of all channels - a spatially neutral signal.

Re-referencing in EEG data analysis: raw recordings with a single reference often contain significant distortions

→ Goal: estimate a more neutral reference point to reduce these effects

Channel Rejection:

Some Channels might have low quality through the entire recording, and should not be directly included in the analyses

Artifacts – What Are They?

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Ocular

- Blinks: large spikes at Fp1/Fp2, symmetric frontal topography, power <5 Hz
- Saccades: step-function time series, left/right asymmetric frontal map
- → ICA blink component is almost always the most obvious one

Cardiac

- Clear QRS complex in time series at ~1 Hz
- Nearly linear gradient scalp topography (heart far from head)
- No spectral peaks — rhythmic pulses across temporal channel

Muscular

- Broadband high-frequency power (20 Hz+) in power spectrum
- Shallow, focal scalp topography (not inside skull)
- Non-stationary bursts — coughing, jaw clenching, head movement

Technical

- Single-channel focal topography (one electrode popped or loose)
- 1/f power spectrum — unlike muscle components
- 50/60 Hz sharp peak = line noise → notch filter or exclude channel

Brain components: dipolar scalp map + 1/f power spectrum with a peak between 5–30 Hz (often 10 Hz alpha) + low dipole residual variance (<15%) — KEEP these!

ICA – Independent Component Analysis

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1

Fit on 1 Hz HP copy

2

Find Bad Components

3

Preview & apply

ICA decomposes the EEG signal into **statistically independent components**. Artifacts (blinks, heartbeat) tend to dominate certain components. We identify and remove them, then reconstruct the clean signal.

What Is an Epoch?

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An epoch is a **fixed-length segment** of EEG data, cut from the continuous recording, time-locked to a specific **event of interest**.

Instead of analysing hours of raw signal, we focus only on what happens around each stimulus — enabling ERP analysis (as covered in the EEG intro!).

Key Parameters

`tmin`	Start relative to event (e.g. −0.1 s)
`tmax`	End relative to event (e.g. 0.6 s)
`event_id`	Which events to epoch (Go / NoGo)
`baseline`	Interval for correction (next slide!)
`reject`	Threshold for auto-rejecting bad epochs

Baseline Correction

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Why do we need it?

EEG signals have slow drifts and DC offsets that vary across trials. Baseline correction removes pre-stimulus activity so that post-stimulus changes reflect genuine neural responses.

How it works

1. Define baseline window (e.g. −100 to 0 ms)
2. Compute mean amplitude in that window
3. Subtract that mean from the entire epoch

→ At t = 0, all channels start at ~0 µV
→ Filtering continuous EEG first prevents filter edge artifacts from contaminating the baseline (erpinfo.org)

Key Takeaways

Re-reference to average before ICA and epoching — removes spatial bias

ICA: fit on 1 Hz HP copy of ORIGINAL raw, apply on Day 1 filtered data

Epochs cut continuous EEG into event-locked windows (ERP trials)

Baseline correction removes pre-stimulus drift — all epochs start at ~0 µV

Time Domain: Event-related potentials

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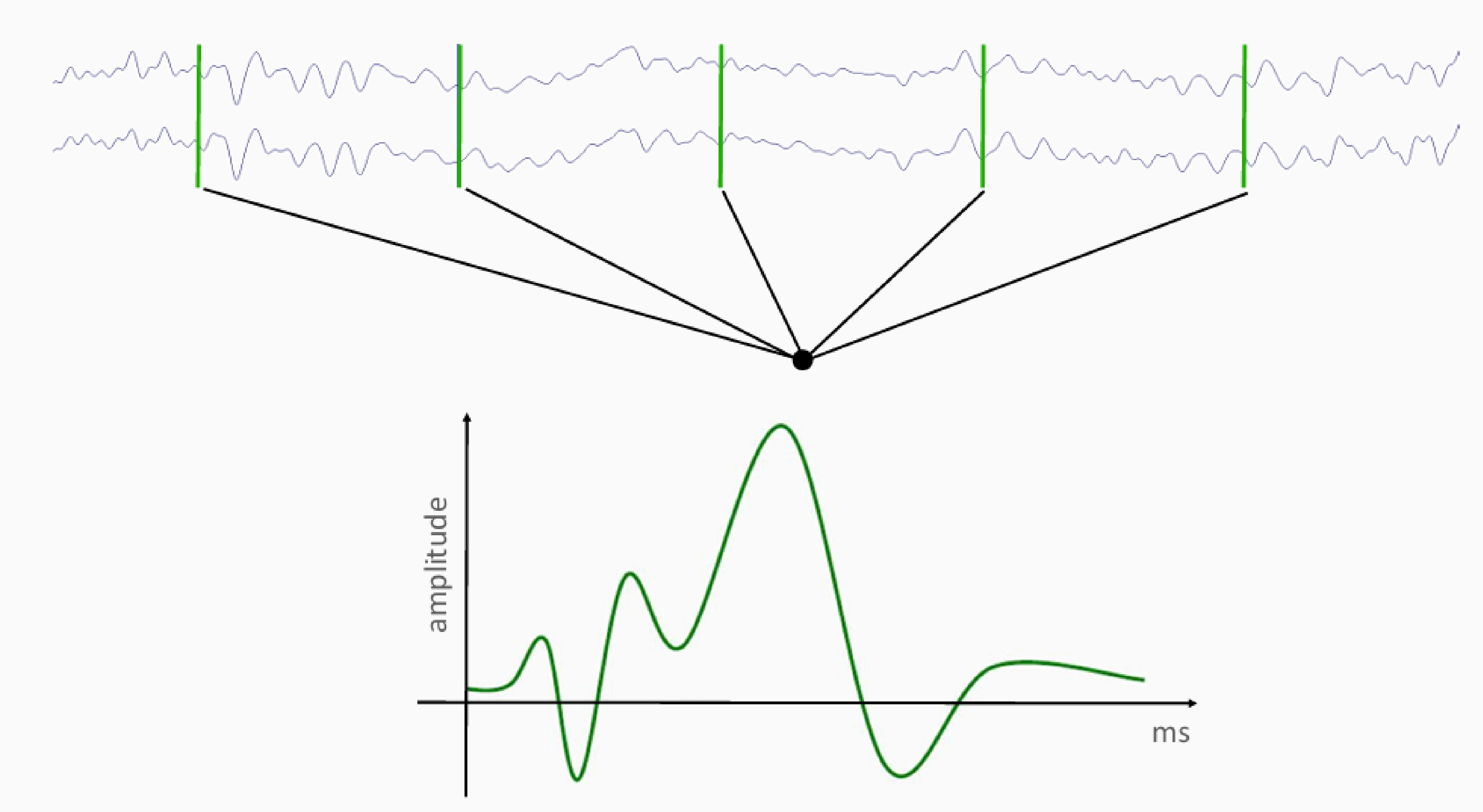
What is an ERP?

... an **electrical potential** related to a specific **event** (e.g. stimulus presentation)

- **mean** of EEG signal for a specified (event-related) time window => based on one/many channels, trials and participants

=> data averaging over many Trials for every epoch!



Trigger/Marker

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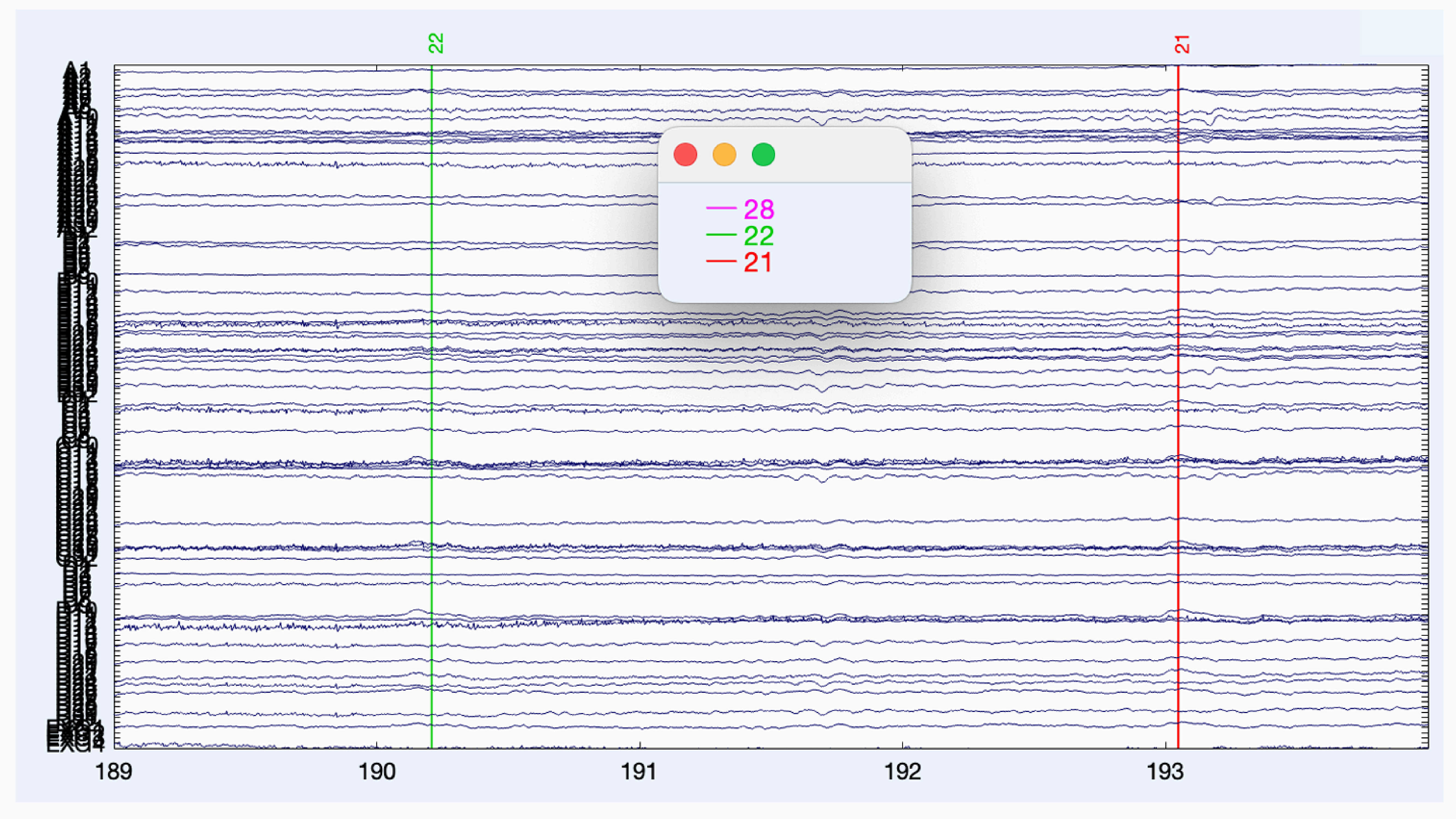

trigger/marker =
time stamp for ERP
analysis

ERP naming system

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Naming system:

- N = negative and P = positive
- number = ms of peak latency after stimulus

  or position of the peak in chronological order of the waveform
- sometimes functional names

Luck, S. J. (2014). Overview of Common ERP Components. In S. J. Luck (Ed.), *An Introduction to the Event-Related Potential Technique* (2nd ed., pp. 71 -118). Bradford Books.

Paradigm

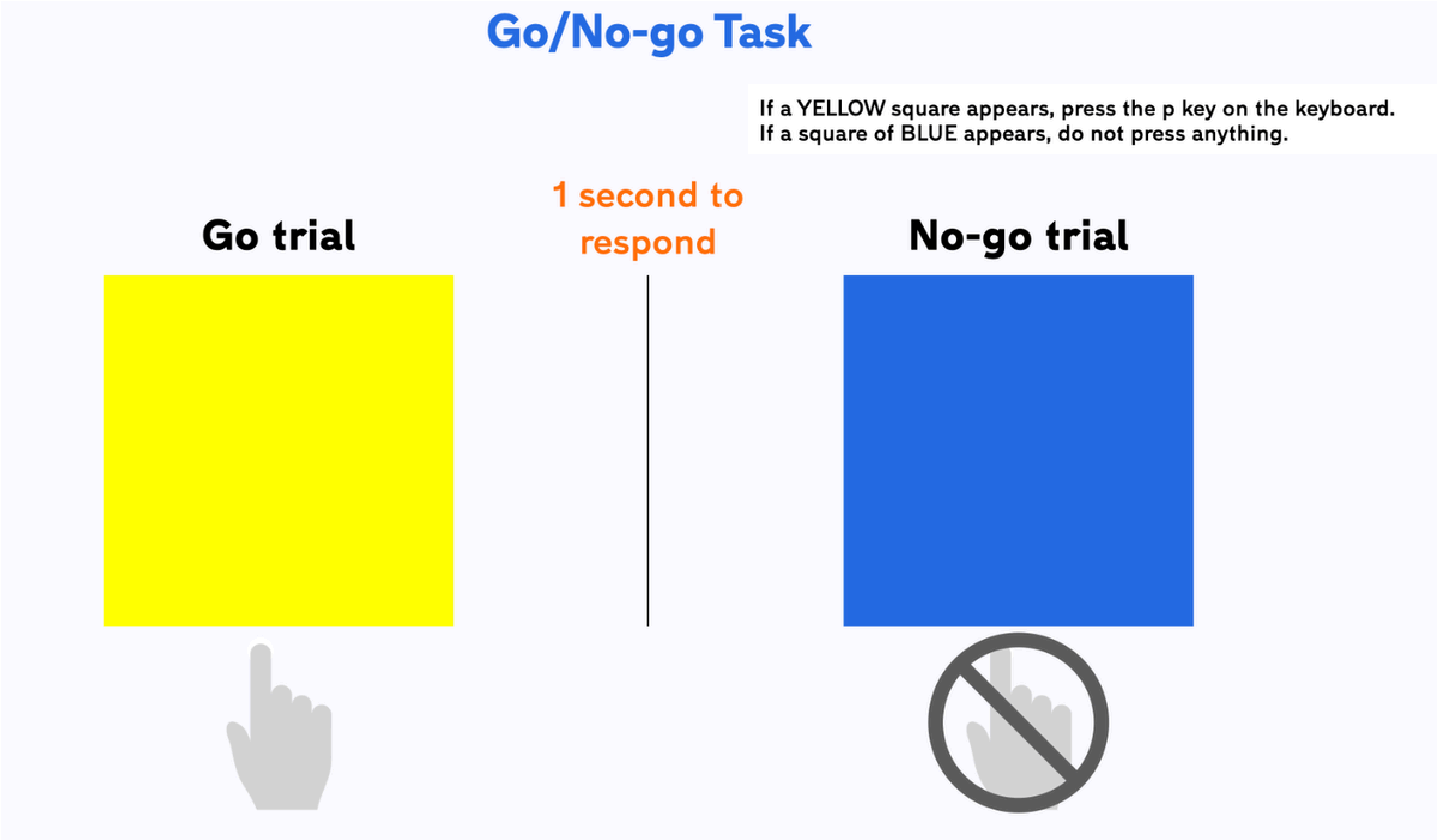
- participants get instructions
- in every trial presentation (e.g. 1s) of either...

  yellow square => press button in time as fast as possible (e.g. 1s to respond)

  or...

  blue square => no response
- most trials are go trials => create an automatic response behaviour
- occasional no-go trials => challenging response inhibition
- important: relative likelihood of go and no-go trials

https://www.testable.org/experiment-guides/executive-function/go-no-go-task



https://www.testable.org/wp-content/uploads/2022/11/Go_nogo_illustration.png

Go/No-go task and EEG

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frontocentral ERPs in no-go trials:

- no-go N200 → conflict detection / response monitoring
- no-go P300 → response inhibition / evaluation of action

Typical characteristics:

- strongest over frontocentral electrodes (e.g. Fz, Cz)
- larger amplitudes in No-Go vs Go trials

Component Analysis

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Peak Amplitude & Latency

→ Peak amplitude: the maximum (or minimum) voltage value of a component within a defined time window.

→ Peak latency: the time point (ms) at which the peak occurs after stimulus onset.

→ Advantage: easy to compute & interpret.

→ Disadvantage: sensitive to noise — a single outlier trial can shift the peak.

Mean Amplitude

→ Average voltage across all time points within a predefined window (e.g., 200–350 ms for N2).

→ Advantage: more robust to noise and small waveform distortions.

→ Disadvantage: requires an a priori window selection based on prior literature.

→ Recommended for most ERP analyses (Luck, 2014, Ch. 9).

Luck, S. J. (2014). An Introduction to the Event-Related Potential Technique (2nd ed.), Ch. 9. Bradford Books.

💡 *In practice: inspect waveforms visually first, then select the method that best matches your research question.*

ERPs - Go-/No-go task

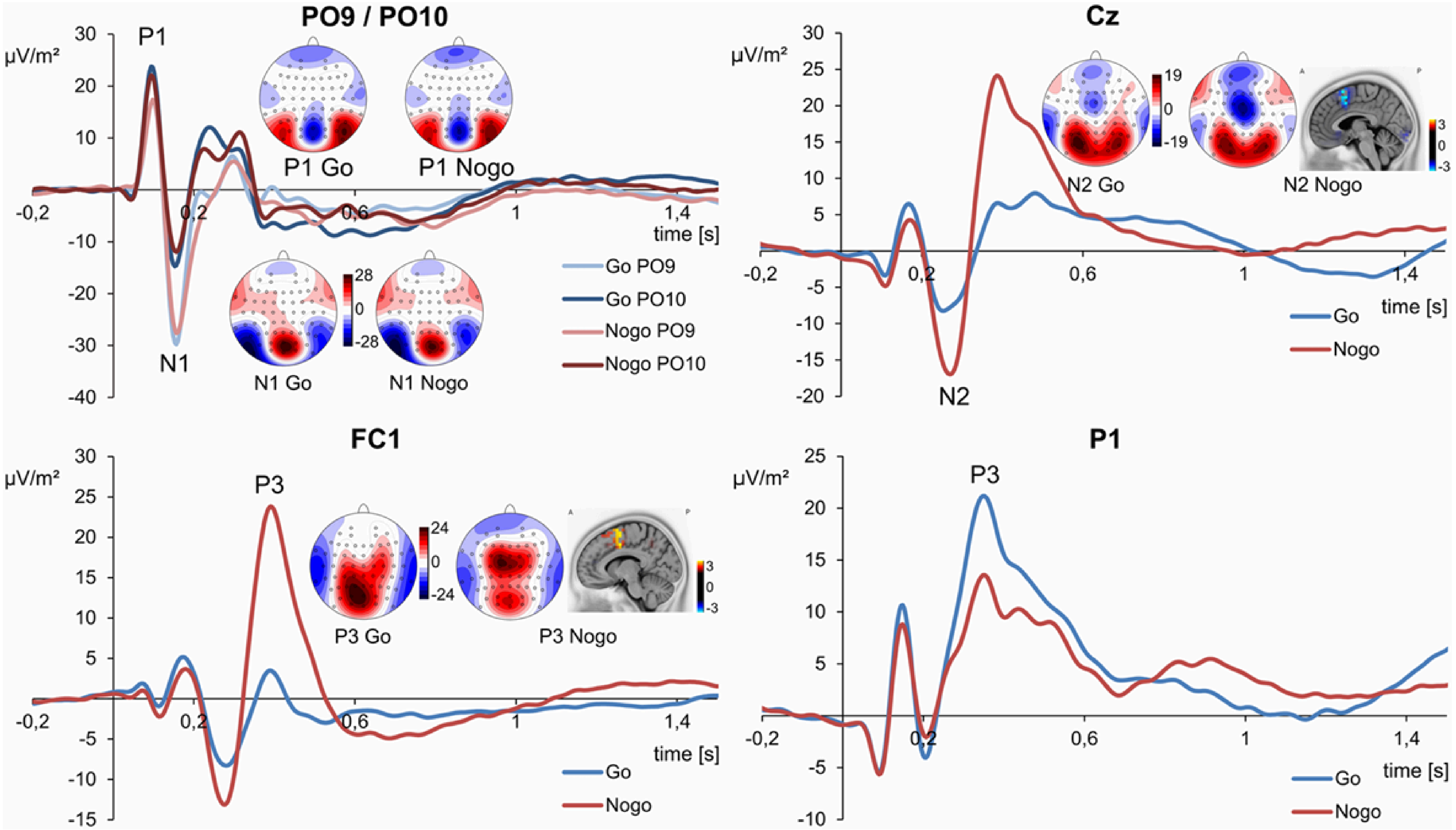

Figure 1. Event-related potential (ERP) components on Go and Nogo stimuli presentations. Plots are given for electrodes PO9/PO10 depicting P1 and N1 ERP, electrode Cz for N2, electrode FC1 for Nogo-P3 as well as electrode P1 for Go-P3. ERPs on Go stimuli are shown in blue, ERPs on Nogo stimuli are shown in red. The scalp topography plots reveal the distribution of voltages at the time point of the peak maximum of each ERP component. Time point zero denotes the time point of stimulus delivery. For N2 and P3 ERP, the results of sLORETA source estimation are shown.

Vahid, A., Mückschel, M., Neuhaus, A., Stock, A. K., & Beste, C. (2018). Machine learning provides novel neurophysiological features that predict performance to inhibit automated responses. Scientific reports, 8(1), 16235.

Interpretation – Relation Behavior and EEG

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Statistical Approach

Correlation analysis

Pearson/Spearman r between ERP amplitude and behavioral measure (e.g., d' vs. P3)

ANOVA / t-test

Compare ERP amplitudes for correct vs. incorrect trials

Regression models

Predict behavioral outcome from ERP component features

Caution:

Correlation ≠ causation; consider multiple comparisons correction

→ *Reference: Luck, S. J. (2014), Ch. 10 – Statistical Analysis of ERP Data*

An Outlook to more Complex Analysis methods

- For ERPs: Mass Univariate Analysis
- Time-Frequency Analysis
- Microstate Analysis
- Source Localization

chapter 9 in : Luck, S. J. (2014). An Introduction to the Event-Related Potential Technique (2nd ed.). Bradford Books

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Any other questions?

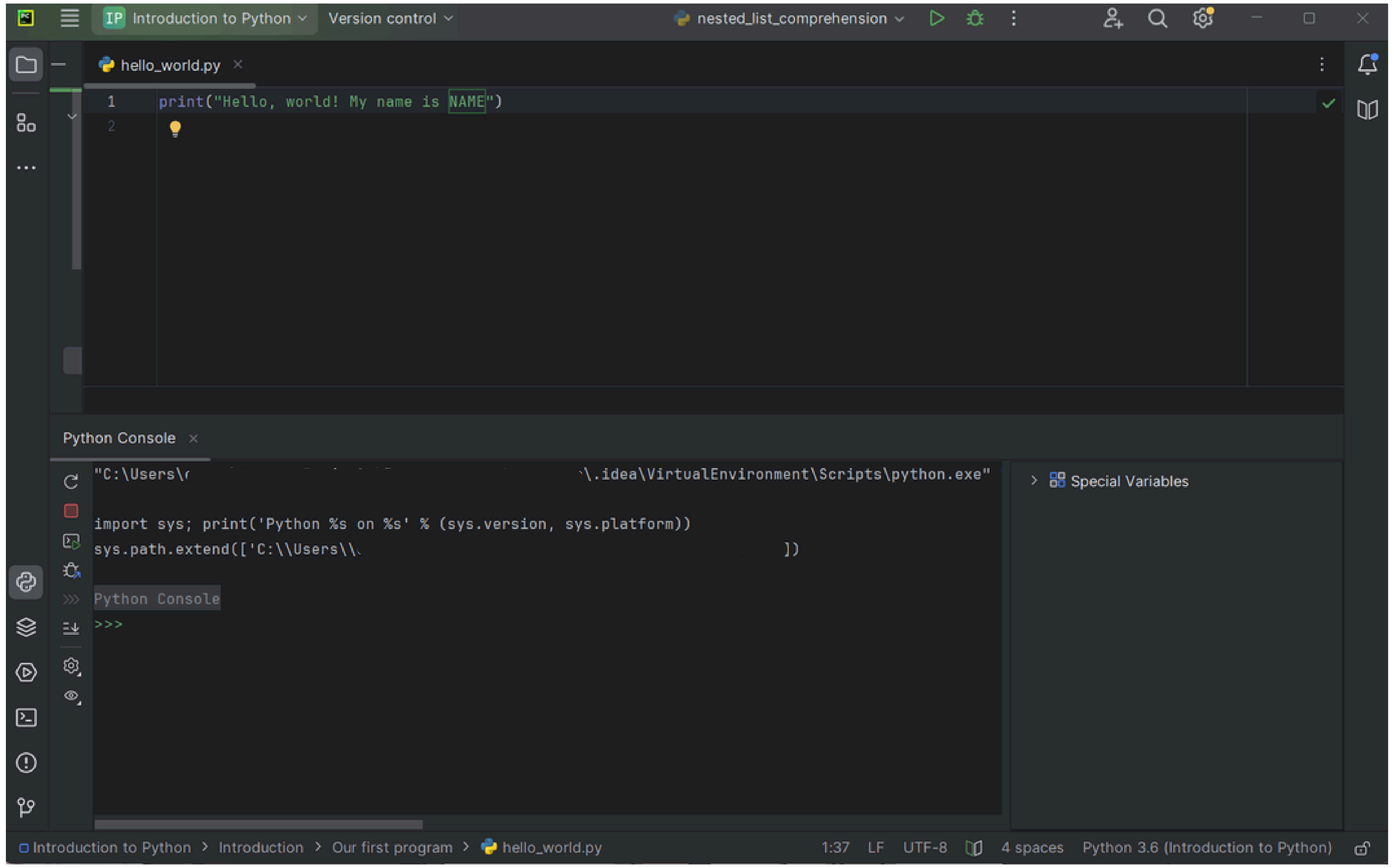
Thank you for your attention!

More About SNN

Programmes

Python

- open-source coding language
- object-oriented, high-level programming language with dynamic semantics
- widely used in web applications, software development, data science, and machine learning (ML)
- quick and easy to learn and understand
- further introduction to Python is in the GitHub repository



Programmes

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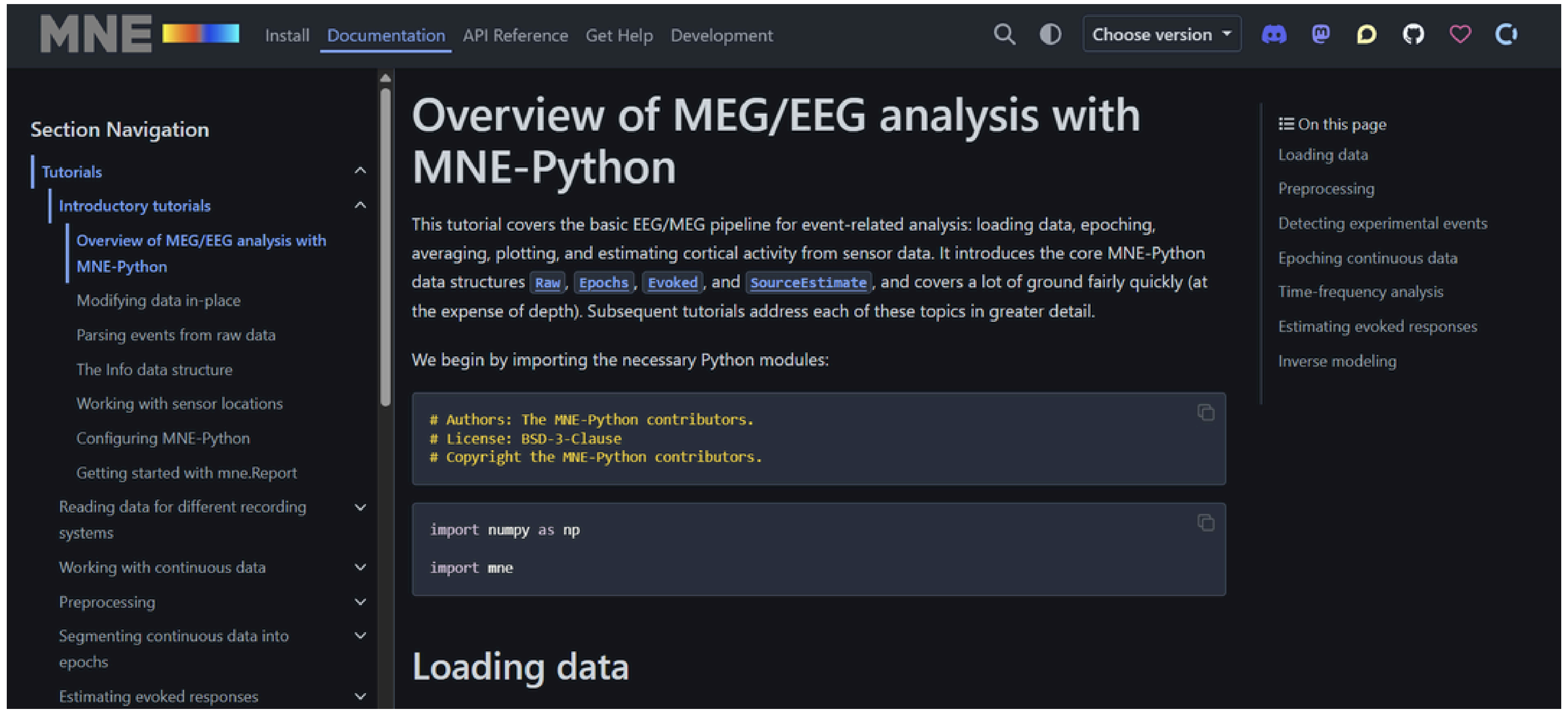
MNE

- open-source Python software
- processing, analyzing, and visualizing electroencephalography (EEG) and magnetoencephalography (MEG) data
- comprehensive tools for the full analysis pipeline
    - pre-processing (filtering, artifact removal)
    - sensor-level analysis
    - source estimation
    - time-frequency analysis
    - even machine learning approaches (encoding/decoding)
    - visualization and plotting



Links / QR Codes you will need

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GitHub Repository with slides & code: https://github.com/jujumistral/EEG_hackathon_SNN

Download Python, an IDE & Python MNE: ForParticipants/Roadmap.pdf **in github**

Data: https://tinyurl.com/eeg-data-psyfako

Literature and Sources

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- slides Merle Sagehorn (University of Osnabrück)
- https://www.testable.org/experiment-guides/executive-function/go-no-go-task

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